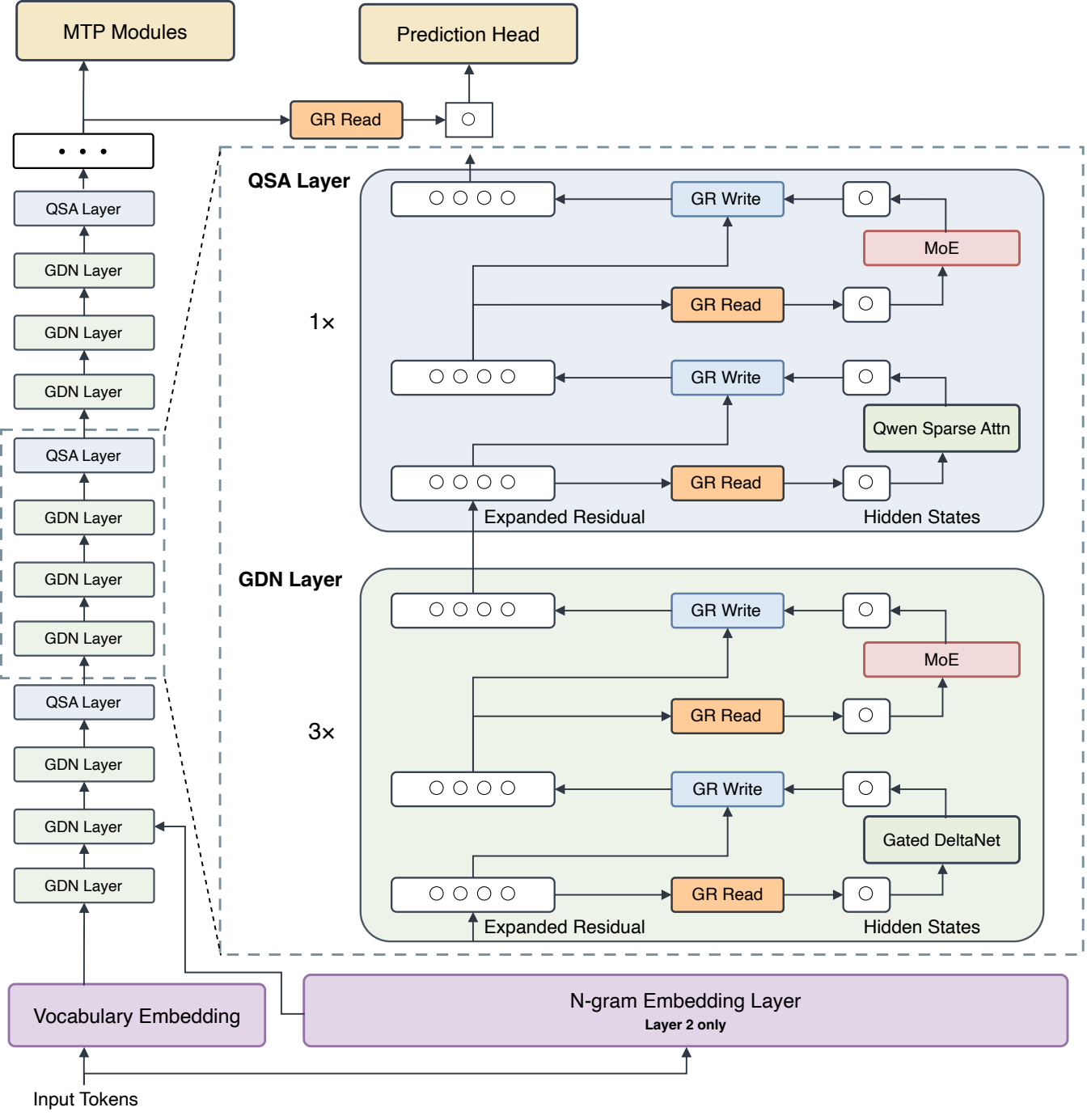

MTP Modules
Prediction Head
GR Read
QSA Layer
GDN Layer
GDN Layer
GDN Layer
QSA Layer
GDN Layer
GDN Layer
GDN Layer
QSA Layer
GDN Layer
GDN Layer
GDN Layer
QSA Layer
1×
GR Write
MoE
GR Read
GR Write
Qwen Sparse Attn
GR Read
Expanded Residual
Hidden States
GDN Layer
3×
GR Write
MoE
GR Read
GR Write
Gated DeltaNet
GR Read
Expanded Residual
Hidden States
Vocabulary Embedding
N-gram Embedding Layer
Layer 2 only
Input Tokens